English Interjections as a Homeostatic Property Cluster

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Abstract

English interjections have been denied category status (Jespersen, 1924), dismissed as uninteresting (Huddleston & Pullum, 2005), or acknowledged as "marginal and anomalous" (Quirk et al., 1985, p. 67). This paper argues that interjections are a thin-but-genuine homeostatic property cluster (HPC): a set of co-occurring properties (prosodic isolation, non-inflection, syntactic non-integration, non-referentiality, emotive force) maintained by identifiable stabilisers. At least three diachronic pathways – grammaticalization via semantic bleaching (historically demonstrated), pragmaticalization via syntactic deactivation, and nonce coinage via onomatopoeia (both proposed with clear examples) – converge on the same synchronic cluster, extending Gehweiler (2008)'s analysis of *gee!* to a general theoretical claim. A perturbation probe suggests that removing a stabiliser produces predictable, asymmetric category shifts. The paper's distinctive contribution is that interjection projectibility is primarily pragmatic rather than truth-conditional: the most frequent interjections are interactional tools (continuers, repair initiators, change-of-state tokens), and categorizing a novel form as an interjection licenses inferences about prosodic behaviour, interactional distribution, and stance expression. A field-relative decomposition shows that syntacticians, semanticists, and interactional linguists each track distinct HPCs with overlapping but non-identical extensions. The category that embarrassed classical theory is a natural fit for HPC – and its pragmatic projectibility extends the framework to a new domain of inference.

1 Introduction

In 1586, William Bullokar wrote the earliest grammar of English. It included interjections. Bullokar defined them as words that betoken "a sudden passion of the mind" (Bullokar, 1586/1980, p. 373) – a definition that foregrounds meaning and use rather than distributional properties.

Four centuries later, the category's status has declined. Jespersen (1924, p. 90) treated interjections not as a lexical category but as a manner in which words of other categories may be used. Quirk et al. (1985, p. 67) included the category but called it "marginal and anomalous." Huddleston and Pullum (2002, p. 22) listed interjection as one of nine lexical categories in their comprehensive grammar (*ah*, *damn*, *gosh*, *hey*, *oh*, *ooh*, *ouch*, *whoa*, *wow*), but Huddleston and Pullum (2005, p. 16) dropped it from

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the textbook because "there really isn't anything interesting for a grammar to say" about interjections. A category real enough for the definitive grammar but thin enough for the textbook to cut.

The empirical picture, though, hasn't simplified. English interjections span phonology, morphology, syntax, semantics, and pragmatics. The category resists clean definition – no single property is necessary and sufficient – and its boundaries with nouns, verbs, adverbs, fillers, and routine formulae are graded, not sharp. Cross-linguistically, "interjection" is at best a comparative concept (Haspelmath, 2010). This paper asks not "are interjections a category?" but "what does categorizing something as an interjection let you predict?" A category that supports no inferences is a filing label. A category that supports reliable inferences across novel instances is a genuine kind.

A methodological note: HPC membership, as used here, attaches to constructions (uses), not to lexemes. *Damn* as a standalone interjection, *damn* as an attributive modifier in *the damn car*, and *damn* as a verb in *I damn you* are different constructions with different cluster profiles – they may belong to different HPCs or to the same HPC in different degrees. Lexemes serve as convenient labels; the unit of analysis is the construction.

Section 2 documents the cluster. Section 3 introduces the theoretical framework – homeostatic property cluster (HPC) theory (Boyd, 1999) – and explains why it handles the interjection data better than classical or prototype-based alternatives. The rest of the paper identifies the mechanisms that sustain the cluster, extending Gehweiler (2008)'s single-case analysis (published in this journal) to a general theoretical claim, and argues that interjections exhibit a distinctive form of projectibility – primarily pragmatic rather than truth-conditional.

2 The property cluster

English interjections exhibit a cluster of co-occurring properties across phonology, morphology, syntax, semantics, and pragmatics. This section surveys those properties, drawing primarily on the empirical base documented in Ameka (1992), Quirk et al. (1985), and Huddleston and Pullum (2002). For each property, two questions are asked: does it co-occur with the others, and does observing it in a novel form license inference to the rest of the cluster?

2.1 Phonology

Interjections are prosodically isolated. They typically occupy their own intonation unit, separated from surrounding discourse by pauses (Ameka, 1992, p. 108). In writing, this isolation surfaces as punctuation – commas, dashes, exclamation marks – setting the interjection off from the clause it accompanies (Huddleston & Pullum, 2002, p. 1350).

Interjections may also contain phonemes not otherwise found in the speaker's dialect. The interjection *uh-oh* ([əʔo]) is one of the few cases where a glottal stop appears in dialects that otherwise lack it. Other atypical sounds include the alveolar clicks in *tut-tut* ([ǀǀ]), the voiceless bilabial fricative in *whew* ([ɸɪu]), and – for dialects that no longer use it – the voiceless velar fricative in *ugh* ([əx]) (Quirk et al., 1985, p. 853). Spelling pronunciations often arise for these forms (e.g., [tət tət] for *tut-tut*, [əg] for *ugh*), suggesting that the atypical phonemes are marked even for native speakers.

Inferential contribution. Prosodic isolation is a strong cue: if a novel form occupies its own intonation unit and resists integration into the surrounding prosodic contour, a listener can predict syntactic non-integration and non-referentiality. The atypical phonemes reinforce this inference. But prosodic isolation isn't unique to interjections – vocatives, appositives, and other supplements share it (cf. Wilkins, 1992). What distinguishes interjections is the convergence of prosodic isolation with the properties below.

2.2 Morphology

English interjections characteristically don't inflect or derive. They resist the morphological operations that characterize other open-class categories: no plural (**ouches*), no tense (**ouched*), no comparative (**oucher*), no nominalization (**ouchness*) (Ameka, 1992, p. 106). A caveat: Libert (2019) notes that "few, if any, papers focus on the morphology of interjections", so the apparent absence of morphological productivity may partly reflect a gap in the research rather than a firm empirical generalization.

Some interjections retain remnant morphology from earlier categorial membership. *Heavens* and *bollocks* preserve the nominal plural suffix, and *goddammit* preserves a verb-object-pronoun structure, but none of these remnants are productive. The plural in *heavens* doesn't contrast with a singular **heaven!* (as an interjection), and speakers don't extend the pattern to new forms.

Inferential contribution. Non-inflection predicts non-argumenthood. Forms that don't inflect can't agree with a verb or take a determiner, so they can't fill the syntactic slots that require these operations. But non-inflection alone isn't diagnostic: adverbs also don't inflect (Ameka, 1992, p. 106). The distinction is syntactic: adverbs form constituents with other words, while interjections don't (Meinard, 2015, p. 152).

2.3 Syntax

Interjections characteristically function as SUPPLEMENTS – "parenthetical strings that are not integrated in clause structure" (Huddleston & Pullum, 2002, p. 1350). In *Damn, we're going to be late!*, the interjection isn't a complement or modifier of anything in the clause. It's appended to the beginning as a parenthetical.

Supplement function isn't unique to interjections. Relative clauses, noun phrases, adjective phrases, and adverb phrases can all function as supplements (Huddleston & Pullum, 2002, p. 1356):

(1) a. *Ah*, so you were there after all! (interjection)
b. We called in to see Sue's parents, *which made us rather late*. (relative clause)
c. A university professor, *Dr. Brown*, was arrested. (NP)

What distinguishes interjection supplements is their combination with the other cluster properties: non-referentiality, non-inflection, prosodic isolation, and emotive force. The function is shared; the category isn't.

A handful of interjections exceptionally take complements. Huddleston and Pullum (2002, 1361fn) analyse *damn* in *damn these mosquitoes!* as the head of an interjection phrase (IntjP) taking an NP complement – one of the very few cases where an interjection projects beyond a single word. But even here the syntax is constrained: no subject is present or intended, and there's no tense inflection. This is better treated as a residue of verb syntax than as full verbal behaviour – a point that matters for the recruitment pathways discussed in Section 4. (The internal structure of lexicalized pairs like *oh boy* and *holy cow* is less clear and won't be resolved here.)

Inferential contribution. Supplement function predicts prosodic isolation and resistance to syntactic embedding:

(2) a. * *I think that damn, we're going to be late.*
b. * *Damn and quickly, we left.*

It also predicts a semantic consequence: if the form is syntactically non-integrated, its contribution is independent of the proposition expressed by the host clause. Remove the interjection and the truth conditions don't change – the hallmark of non-truth-conditional meaning (cf. Potts, 2007, p. 166).

2.4 Semantics

English interjections characteristically don't refer. While nouns and verbs typically pick out entities or events in the world, interjections express internal states of their users (Leech, 2006): anger (*damn*), disgust (*eww*, *yuck*), surprise (*wow*), regret (*alas*), pain (*ouch*), realization (*eureka*). Some interjections are more propositional than emotive – *yes* and *no* affirm or deny propositions, and *oh* can signal recognition of new information rather than expressing feeling. Wierzbicka (1992) argues that interjections have genuine semantic content, decomposable via Natural Semantic Metalanguage into deictic primitives (*I*, *you*, *now*, *here*). This is not incompatible with non-referentiality as used here: the claim is that interjections don't pick out entities in the world, not that they're meaningless. Wierzbicka's explications characterize the use-conditional content that interjection$_{\text{sem}}$ (Section 3.4) tracks. Non-referentiality is a cluster property, not a definitional requirement.

The contrast with nouns is sharpest where both share a lexical source. *Jesus* as a noun refers to a person; *Jesus!* as an interjection doesn't. The interjection use has been "bleached of [its] original meaning" – it no longer refers to the entity the noun originally picked out (Gehweiler, 2008, p. 72). This bleaching is part of the recruitment pathway from noun to interjection (Section 4).

Inferential contribution. Non-referentiality predicts distributional restrictions. A form that doesn't refer can't be made definite (**the damn!*), can't be quantified over (**every ouch*), and can't fill argument positions (**Damn surprised me*, **There's damn*). These are negative predictions, but they're non-trivial: they distinguish interjections from nouns that have undergone only partial bleaching.

2.5 Pragmatics

Interjections typically carry exclamatory or imperative force: *ouch!* expresses pain, *shh!* demands silence (Quirk et al., 1985, p. 88). But the pragmatic range extends beyond exclamation. Interjections greet (*hey*, *hello*), mark agreement (*yes*, *amen*), indicate understanding (*oh*, *uh-huh*), and signal backchanneling – the listener's ongoing attention to a speaker's turn (Dingemanse, 2020; Stivers, 2019).

The most common interjections tagged in the Corpus of Global Web-Based English (GloWbE; Davies & Fuchs, 2015) confirm this pragmatic breadth: *yes*, *no*, *oh*, *yeah*, *hi*, *hey*, *wow*, *hello*, *ah*, *ha*. The list is dominated by interactional items – forms whose primary function is to manage the conversation rather than to contribute propositional content. Dingemanse (2024) estimates that roughly one in seven conversational turns is an interjection, and the most frequent are not involuntary exclamations but interactional tools: continuers like *mm-hm* that scaffold extended turns, repair initiators like *huh?* that calibrate mutual understanding, and change-of-state tokens like *oh* that display evolving knowledge (cf. Norrick, 2009).

The boundary with ROUTINE FORMULAE is instructive. Coulmas (1981, pp. 2–3) defines routine formulae as "highly conventionalized prepatterned expressions whose occurrence is tied to more or less standard communication situations": *bye*, *hello*, *sorry*, *thank you*. Ameka (1992, pp. 109–110) argues these differ from interjections in three ways: they have addressees, they're predictable responses to social conventions, and they're always speech acts rather than mere reflections of the speaker's mental state. Wilkins (1992, p. 142) counters that these differences make routine formulae "a distinct pragmatic and semantic subtype of interjections", not a separate category.

Routine formulae share some properties with interjections (prosodic isolation, pragmatic function) but have additional characteristics (social convention, addressee-directedness) that interjections don't require. They sit at the boundary – a pattern that will turn out to be theoretically significant (Section 6.3).

Inferential contribution. If a novel form carries emotive or interpersonal force without contributing propositional content, you can predict specific interactional behaviour: turn-initial or standalone positioning, availability for backchanneling, and resistance to embedding under reported speech (**She said that damn*). These are predictions about distribution and sequential position, not restatements of the meaning type. They follow because the mechanisms that maintain emotive force (pragmatic conventionalization, social indexing) also maintain the interactional niche.

2.6 Co-occurrence, not necessity

These properties co-occur statistically, not necessarily. Every characterization of interjections is hedged. Descriptions across the literature characteristically hedge: interjections "commonly" occupy their own intonation unit, "typically" lack inflection, "usually" resist syntactic integration, and "generally" express emotive meaning (cf. Ameka, 1992; Wilkins, 1992). No single property is necessary and sufficient.

The pattern calls for explanation. The cluster holds together not because a definition mandates it but because processes couple the properties: prosodic isolation reinforces syntactic non-integration, non-referentiality removes the need for argument structure, and pragmatic conventionalization entrenches the emotive function.

The coupling is visible when it breaks. *Damn* as an interjection (*Damn!*) has the full cluster: prosodic isolation, syntactic non-integration, non-referentiality, emotive force. *Damn* as an attributive modifier (*the damn car*) retains emotive force but loses prosodic isolation and syntactic non-integration – it's embedded inside an NP (cf. Potts, 2007). The properties don't erode randomly; they erode together when the mechanisms that couple them (here, prosodic-syntactic independence) are neutralized. A list of co-occurring properties can't explain this pattern. A cluster whose co-occurrence is maintained by identifiable processes can.

Table 1 makes the cluster inspectable. Each row is a construction (not a lexeme; see Section 1), and the columns show its profile against the five cluster properties and three field-relative HPCs from Section 3.4. The table reveals where the HPCs converge (standalone *damn!*, *oh*) and where they decouple (attributive *the damn car* is interjection$_{\text{sem}}$ but not interjection$_{\text{syn}}$; *mm-hm* is interjection$_{\text{int}}$ but debatably interjection$_{\text{syn}}$). Boundary items (*bye*, *um*) share some but not all cluster properties – the pattern that Section 6.3 explains as mechanism overlap.

The next two sections develop the theoretical framework (Section 3) and the mechanisms themselves (Section 4).

3 HPC theory and why projectibility matters

The property cluster documented above has a distinctive shape: statistically reliable co-occurrence, no necessary and sufficient conditions, graded boundaries, and hedged universals. Classical category theory can't accommodate this. HPC theory can.

Table 1: Diagnostic matrix: constructions against cluster properties and field-relative HPCs. ✓ = property present; (✓) = partial or debated; — = absent. Sources: properties from Ameka (1992), Quirk et al. (1985), Huddleston and Pullum (2002); interjection$_{sem}$ from Potts (2007); interjection$_{int}$ from Stivers (2019), Dingemanse (2020).

	Prosodic isol.	*Non-inflecting*	*Synt. non-int.*	*Non-referential*	*Emotive/stance*	*Intj$_{syn}$*	*Intj$_{sem}$*	*Intj$_{int}$*
gee!	✓	✓	✓	✓	✓	✓	✓	✓
damn!	✓	✓	✓	✓	✓	✓	✓	✓
damn these ...!	✓	(✓)	(✓)	✓	✓	(✓)	✓	—
the damn car	—	—	—	—	✓	—	✓	—
oh	✓	✓	✓	✓	✓	✓	✓	✓
mm-hm	✓	✓	✓	✓	(✓)	(✓)	—	✓
huh?	✓	✓	✓	✓	(✓)	(✓)	—	✓
bruh	✓	✓	✓	✓	✓	✓	✓	✓
bye	✓	✓	✓	(✓)	—	✓	—	(✓)
um	(✓)	✓	✓	✓	—	(✓)	—	(✓)

3.1 The framework

Boyd (1999) proposed that natural kinds are not defined by essences but sustained by homeostatic mechanisms. A kind is a property cluster: a set of properties that co-occur reliably across instances. The co-occurrence isn't accidental; it's maintained by mechanisms that keep the properties clustered. And the clustering matters because it supports induction – it lets you predict properties of new instances from observed properties of old ones.

Following Boyd (1999), I adopt a three-part slogan: a linguistic category is a profile of co-occurring properties, stabilized by mechanisms, projectible relative to purposes. The first two legs matter because they make the third possible. Projectibility is the payoff: a category just is whatever supports reliable inference across novel instances. The cluster is the visible trace; the inferential role is the substance.

Consider prosodic isolation: if interjections lost it – if they were integrated into the surrounding prosodic contour – would the category hold together? A STABILISER is any process whose removal would change the clustering – would cause the category to fragment, drift, or dissolve (cf. Boyd, 1999). This yields the perturbation probe: if a proposed stabiliser can be neutralized and the cluster remains intact, the stabiliser wasn't doing the work claimed for it.

Not everyone agrees that mechanisms are necessary. Slater (2015) argues that stable property clustering suffices for natural kindhood without homeostatic mechanisms: a stably co-instantiated property cluster (SPC) is projectible because the stability itself underwrites induction, regardless of why the properties co-occur. Khalidi (2013) makes a related argument from causal networks: kinds are nodes in causal structures, and what matters is reliable co-occurrence within the network, not a specific homeostatic mechanism. These alternatives deserve honest engagement, because the present paper's evidential state is closer to SPC than to full HPC: one confirmed stabiliser (prosodic-syntactic coupling), one boundary enforcer that is not homeostatic (the non-referentiality ratchet), and one candidate (social indexing). That is thin for “homeostatic.”

The paper’s strongest contributions – the field-relative decomposition, the pragmatic-kind diagnosis, the boundary predictions – do not require mechanisms. They are available to SPC as well. What does require mechanisms is the perturbation ordering prediction: that the order of property erosion tracks which mechanism is disrupted. This is the distinguishing test between HPC and SPC. If the perturbation evidence holds up under further testing, the mechanism requirement is justified; if it doesn’t, the interjection category is better treated as an SPC with pragmatic projectibility but without homeostatic maintenance. That is a weaker but still substantive claim.

The paper therefore positions itself as demonstrating stable clustering and pragmatic projectibility (on which the evidence is strong) while arguing that the mechanism story is worth pursuing (on which the evidence is preliminary). It succeeds or fails on this question: does categorizing a novel form as an interjection license non-trivial inferences for specific purposes? If it merely lists co-occurring properties, it has described a cluster, not demonstrated a kind.

3.2 HPC vs. prototype theory

Prototype theory (Rosch, 1975; Taylor, 2003) was developed for exactly the kind of data Section 2 documents, and it handles it well. Graded membership, better-and-worse exemplars, fuzzy boundaries via overlapping similarity spaces – all of these are core predictions of prototype theory, and all are confirmed by the interjection data. Prototype theory also correctly predicts that boundary disputes (interjections vs. fillers, vs. routine formulae) should be persistent rather than resolvable by definition. These are genuine explanatory achievements, and HPC doesn’t replace them.

HPC adds a different kind of question. Prototype theory describes the fact that properties cluster but not why they cluster in the pattern they do. It generates no predictions about the order in which properties erode when items leave the category (asymmetric fraying), whether apparent gradience decomposes under social conditioning, or whether one property is more central to cluster stability than others (hub-node asymmetry). These predictions are examined for interjections in Section 6.

The perturbation probe provides a further discriminator. The interjection-as-HPC claim is falsifiable: if no proposed stabiliser’s removal would change the clustering, the “homeostatic” part is empty. Prototype theory offers no analogous test.

3.3 What HPC adds beyond property lists and features

Three analytical stances generate different predictions for interjections. A *property list* (descriptive clustering) says that interjections share prosodic isolation, non-inflection, non-referentiality, and emotive force, but makes no prediction about what happens when one property changes. A *feature specification* (e.g., [+PERIPHERAL, -ARGUMENT-TAKING, +EXPRESSIVE]) generates some of the same distributional consequences as HPC (no inflection follows from no phi-features) but treats these as entailments of the feature bundle, not as maintained by identifiable processes. HPC adds three things that neither alternative provides:

1. Perturbation ordering. When a property is disrupted (e.g., morphology gain), HPC predicts that the order of erosion tracks which mechanism is disrupted. Feature theory predicts property loss but not a specific order; a property list predicts nothing.
2. Convergent recruitment. Multiple diachronic pathways converge on the same synchronic cluster because they are constrained by the same stabilisers. Feature theory says nothing about why the features co-occur; a property list simply records that they do.

3. Field-relative decomposition. Different disciplines track overlapping but non-identical HPCs maintained by different mechanisms. Feature theory assigns a single feature bundle; HPC allows one form to participate in multiple overlapping kinds for different purposes.

If a feature-based account can replicate the perturbation ordering and the field-relative decomposition, HPC is doing no additional explanatory work. The empirical test is whether the predictions in Section 6 succeed or fail.

3.4 Field-relative projectibility

Boyd (1999, p. 150) argued that the naturalness of a kind is relative to a disciplinary matrix: "A kind may be natural 'from the point of view of' some discipline or disciplinary matrix but not 'from the point of view of' another." His example is jade, a natural kind in gemology but not in geology (jadeite and nephrite are chemically distinct). In linguistics, the parallel is the noun/name distinction: syntacticians track NOUN (heads NPs, takes determiners, inflects for number); semanticists track NAME (picks out individuals, resists descriptive modification, creates referential opacity). Same words, different HPCs, different mechanisms, different projectibility.

This matters for interjections. Multiple fields recognize a category whose extension is roughly the same set of lexemes, but each field tracks different properties, maintained by different mechanisms, projectible for different purposes. I distinguish these with subscripts:[1]

- Interjection$_{\text{syn}}$ – the syntactician's category (*CGEL*, Quirk et al.). Distributional cluster: supplement function, non-inflecting, non-argument-taking. Maintained by prosodic-syntactic coupling. Projects syntactic behaviour.
- Interjection$_{\text{sem}}$ – the semanticist's category, overlapping with Potts (2007)'s EXPRESSIVES. Use-conditional meaning cluster: independent of at-issue content, nondisplaceable, perspective-dependent, descriptively ineffable (Potts, 2007, p. 166). Maintained by semantic bleaching. Projects meaning type. Different extension: *damn* in *the damn car* is interjection$_{\text{sem}}$ but not interjection$_{\text{syn}}$.
- Interjection$_{\text{int}}$ – the interactional linguist's category, overlapping with Stivers (2019)'s response tokens and Dingemanse (2020)'s liminal signs. Sequential/stance cluster: turn-initial, backchannel, stance display. Maintained by conversational routines and possibly by social indexing. Projects interactional distribution. Different extension: *mm-hm* is interjection$_{\text{int}}$ but debatably interjection$_{\text{syn}}$.

The subscript notation makes explicit that these are overlapping but non-identical HPCs, each maintained by different mechanisms and projectible for different purposes. The traditional label "interjection" names the intersection, and the paper's remaining sections analyze that intersection from multiple angles.

4 Stabilisers: recruitment and maintenance

The property cluster documented in Section 2 raises two questions that co-occurrence alone can't answer: how do items enter the cluster, and what keeps it coherent once they're in it? In what follows, "the interjection cluster" refers to the intersection of the three HPCs introduced in Section 3.4 –

[1] An analogous noun/name analysis uses different words for the syntactic and semantic HPCs because both terms are established in their respective traditions. For interjections, every field uses the same word, so subscripts are the honest notation. An alternative would have been to adopt distinct terms from the relevant literatures – EXPRESSIVE from Potts (2007) for the semantic HPC, RESPONSE TOKEN from Stivers (2019) or LIMINAL SIGN from Dingemanse (2020) for the interactional HPC. The Latin EJACULATIO ("a throwing out", mirroring INTERJECTIO, "a throwing between") would have been etymologically perfect for the interactional category, but modern English has rendered the term unavailable.

the property profile shared by forms that belong to interjection$_{syn}$, interjection$_{sem}$, and interjection$_{int}$ simultaneously. Boyd (1999)'s framework requires both a diachronic account of recruitment and a synchronic account of maintenance.

4.1 Recruitment pathways

Section 2 already pointed to three recruitment pathways: semantic bleaching from nouns, syntactic deactivation from verbs, and nonce coinage via onomatopoeia. The first is historically demonstrated; the second and third are proposed pathways with clear examples but without the extended diachronic documentation that Path 1 has. The first two are analysed in detail here; the third is taken up in Section 4.4.

4.1.1 Path 1: Religious noun to interjection

Gehweiler (2008) traces a four-stage trajectory from proper name to primary interjection, using the development of *gee* from *Jesus* as the central case:

(3) a. Stage 1: Proper name used as invocation (*Jesus!* addressed to the deity)
  b. Stage 2: Exclamatory use, no longer addressing the deity (*Jesus!* as an expression of surprise)
  c. Stage 3: Phonological modification (*Jeez*, *Gee*)
  d. Stage 4: Primary interjection (*gee!*), fully detached from the source noun

The stages don't install cluster properties one by one; prosodic isolation is already present at stage 1, because a vocative like *Jesus!* occupies its own intonation unit (contrast the integrated argument in *Put your faith in Jesus*). What changes is the content: non-referentiality arrives at stage 2, when the speaker is no longer addressing the deity. Phonological divergence at stage 3 makes the etymological connection opaque. By stage 4, the form is a standalone interjection that many speakers don't connect to its source. The vocative is the gateway: it provides the prosodic isolation that the bleaching pathway exploits. The prosodic slot was there first; the semantic content drained out of it (Gehweiler, 2008, p. 85).

This is a constrained trajectory, not a random walk. Each stage makes the next more probable: once the invocation loses its referential force (stage 2), the phonological connection to the source is the last link to the nominal category, and weakening that link (stage 3) enables full independence (stage 4). Gehweiler's corpus data confirms the constraint: the phonological modifications (*Jeez*, *Gee*, *Gosh*, *Golly*) cluster in the seventeenth and eighteenth centuries, with the transition from invocation to primary interjection happening relatively rapidly once the modifications appear (Gehweiler, 2008, pp. 80–85).

Two sub-mechanisms operate within this path. Gehweiler (2008, p. 76) distinguishes gradual Invited Inferencing (speaker-based: the expressive use is conversationally implicated, then conventionalized) from abrupt reanalysis (hearer-based: the hearer parses the exclamation as a standalone unit rather than an invocation). Both feed the same cluster, but they operate at different timescales and involve different agents. That two distinct cognitive processes converge on the same cluster profile is itself evidence for HPC: the cluster can be reached by different routes.

The constrained trajectory generates a specific prediction: partial bleaching (stage 2) should be detectable by the absence of downstream properties. A form at stage 2 should be non-referential but should retain the source phonology and resist full prosodic independence. This is testable and goes beyond historical description.

4.1.2 Path 2: Verb to interjection

A second trajectory enters the cluster via syntactic deactivation rather than phonological modification. *Damn* originates as a verb but in its interjection use has lost subject and tense while retaining – in a few constructions – an NP complement (*damn these mosquitoes!*). The complement retention shows that the bleaching is partial: the form carries some verb properties into the interjection cluster. In *CGEL*'s terms, it's the head of an IntjP, not a VP (Huddleston & Pullum, 2002, 1361fn). The process involves decategorialization: loss of verbal morphosyntax (subject, tense) with gain of expressive force. *Damn* as a verb is a speech act (content meaning: condemning), while *damn!* as interjection$_{\text{syn}}$ is a stance marker (procedural meaning: frustration).

For other verb-derived interjections, the imperative is the gateway – just as the vocative is for Path 1. Imperatives already lack an overt subject and can stand alone prosodically; the interjection use completes the deactivation. *Come on* and *look* (in their discourse-marker uses) have lost not just subject and tense but also complement-taking ability. They function as standalone interjections with no residual argument structure. The contrast is instructive: *damn* represents partial syntactic deactivation (complement retained, subject and tense lost), while *come on* and *look* represent full deactivation (no complement, no argument structure at all). The degree of deactivation correlates with the degree of integration into the interjection cluster. This correlation generates a graded prediction: a form that retains complement-taking ability should also retain other verb-like properties (e.g., prosodic integration in some constructions), while a fully deactivated form should show the complete interjection profile. The mechanism doesn't just describe entry; it predicts the inferential package available at each stage.

4.1.3 Convergent pathways

The two paths enter the cluster from different directions – one through phonological modification of a bleached noun, the other through syntactic deactivation of a verb – but both exploit a gateway construction that already provides prosodic isolation: the vocative for Path 1, the imperative for Path 2. And both converge on the same property profile. The convergence is what HPC predicts: different mechanisms can recruit items into the same kind, producing the same cluster of co-occurring properties through different routes.

Both paths can be situated within Traugott and Dasher (2002)'s subjectification trajectory (content > procedural, nonsubjective > intersubjective), but they instantiate it differently. Path 1 is grammaticalization in Traugott's technical sense: stage 1 (invocation) carries content meaning (addressing an entity) plus subjective stance (reverence); stage 2 (exclamation) loses content and gains procedural meaning (expressive force); stages 3–4 (phonological modification) complete the shift to fully procedural, intersubjective meaning – hearers interpret the form as a conventional expression, not an invocation. Path 2, as described above, involves decategorialization: loss of verbal morphosyntax with gain of expressive force.

Onomatopoeia (Section 4.4) is categorically different: nonce coinages enter the cluster with no prior lexical history – no bleaching, no deactivation, no subjectification. Three diachronically distinct processes (grammaticalization, pragmaticalization, nonce coinage) converge on the same synchronic category. Grammaticalization theory alone can't subsume all three. HPC theory can: what unifies the pathways isn't the diachronic process but the cluster itself – convergent outcomes from different recruitment mechanisms, maintained by the same synchronic stabilisers.

4.2 Synchronic maintenance

Recruitment explains how items enter the cluster. But what keeps the cluster coherent now? Three synchronic processes are relevant: one confirmed stabiliser, one boundary enforcer, and one candidate.

4.2.1 Prosodic isolation and syntactic non-integration

These two properties reinforce each other, though the evidence is stronger in one direction. Prosodic separation makes syntactic integration harder: a form in its own intonation unit is less available as a complement or modifier. Gehweiler (2008, p. 85) provides direct evidence for this direction: sentence-initial position of invocations contributed to the emergence of the interjection – the prosodic slot preceded and enabled the syntactic independence. The reverse direction (syntactic independence licensing prosodic isolation) is plausible – a form with no argument-structural requirements can stand alone prosodically – but the historical evidence is primarily for the prosody-to-syntax direction. The coupling is primarily prosody-driven, with syntactic reinforcement.

4.2.2 The non-referentiality ratchet (boundary enforcer)

Bleaching removes referential content, which removes the semantic motivation for argument structure, which makes the form syntactically free-standing. This is a one-directional cascade, not a feedback loop: once argument structure is lost, there's no pressure to restore it, because the form now occupies a different functional niche. The ratchet is not a homeostatic stabiliser in Boyd's sense (it doesn't resist perturbation through feedback) but a boundary enforcer: it prevents items from drifting back toward their source category. It explains cluster persistence without maintaining cluster cohesion.

4.2.3 Social indexing (candidate stabiliser)

Regional interjections survive because they do identity work. *Lah* in Singapore and Malaysia, *yaar* in India and Pakistan, *haba* in Nigeria, and *word* in African-American English (Lauture, 2020) persist not because of their propositional content (they have none) but because they mark in-group solidarity. Social indexing is a candidate stabiliser, not yet a confirmed one. The open question is whether it preserves the *cluster* (maintaining property co-occurrence) or merely preserves individual *items* (maintaining inventory). If the former, social indexing is a genuine HPC stabiliser; if the latter, it's a lexical-persistence mechanism that operates independently of the cluster.

A test: when a regional interjection is adopted by out-group speakers as ironic quotation, does it retain its cluster properties or lose them? If out-group *lah* loses prosodic isolation or emotive force while retaining its social-marking function, social indexing is maintaining the inventory, not the cluster. If it retains the full profile, social indexing is genuinely maintaining the HPC. The test is unexecuted. The interjection cluster therefore has one confirmed synchronic stabiliser (prosodic-syntactic coupling), one boundary enforcer that prevents reversal without maintaining cohesion (the non-referentiality ratchet), and one candidate stabiliser (social indexing).

4.3 Perturbation probe

If the stabilisers are genuine, disrupting one should change the clustering. The clearest cases involve interjections that gain verbal morphology. *Wow*, *boo*, *ooh*, and *shoo* are interjections with the full cluster profile: prosodically isolated, non-inflecting, non-referential, syntactically non-integrated. But all four have verb uses – *she wowed the audience*, *the crowd booed*, *they oohed and aahed*, *I shooed them away* (Biese, 1941, pp. 178–214) – and in those uses the cluster comes apart asymmetrically across the three HPCs. Interjection$_{\text{syn}}$ properties erode: the derived verb inflects (*wowed*, *booing*), takes arguments (*wowed the audience*), and integrates syntactically. Interjection$_{\text{int}}$ properties erode: the verb use isn't a backchannel or standalone stance marker. But interjection$_{\text{sem}}$ properties partially persist: *wow* as a verb still carries evaluative meaning (*she wowed them* conveys positive assessment, not a neutral description). The erosion isn't random; it tracks which mechanisms are disrupted by the gain of morphology, and the three HPCs come apart at different rates.

An interactional case shows a different erosion pattern. When *mm-hm* – an interjection$_{\text{int}}$ continuer – is replaced by a full clause (*I agree*, *I'm listening*), the form gains propositional content and loses prosodic independence: it integrates into the turn structure rather than occupying its own intonation unit. Interjection$_{\text{int}}$ properties erode (it's no longer a minimal backchannel), and interjection$_{\text{syn}}$ erodes (it's now a clause, not a supplement). But the interactional *function* persists – the speaker is still doing agreement or attention-marking. The erosion pattern differs from the emotive cases above, because different mechanisms are disrupted: gaining morphology disrupts prosodic-syntactic coupling, while gaining propositional content disrupts the non-referentiality ratchet.

A third pattern comes from obsolescence. *Fie* and *zounds* were live interjections in early modern English; neither is used with genuine expressive force today. But the prosodic shell persists: a modern speaker can still produce *fie!* or *zounds!* as a standalone exclamation, and hearers recognize the prosodic form even if the emotive force is gone. The end state is clear: interjection$_{\text{int}}$ properties are absent (no stance display, no backchanneling), the community-dependent aspect of interjection$_{\text{sem}}$ is absent (emotive force requires a community to sustain it), but interjection$_{\text{syn}}$ properties persist (prosodic isolation, non-referentiality, non-inflection). The mechanism-tracking framework predicts that community-dependent mechanisms (conversational routines, social indexing) should weaken before structural ones (prosodic-syntactic coupling, the non-referentiality ratchet), producing a specific erosion sequence: interjection$_{\text{int}}$ before interjection$_{\text{sem}}$ before interjection$_{\text{syn}}$. A pre-registered test of this prediction coded 616 tokens of *fie* in COHA (1820–2019) for all three properties and fitted Bayesian logistic decline models to estimate the ED50 (the decade at which each property's probability crosses 0.5).[2] The result was negative: all three properties declined together ($\tau_{\text{syn}} \approx 1947$, $\tau_{\text{int}} \approx 1948$, $\tau_{\text{sem}} \approx 1949$; 95% CrIs overlapping), with $\Pr(\tau_{\text{int}} < \tau_{\text{sem}} < \tau_{\text{syn}}) = 0.13$. (Late-decade rebounds in the 1980s and 2000s reflect historical fiction and Shakespeare adaptations where characters use *fie* as a live interjection; the confirmatory model's monotone constraint absorbs these.) The predicted sequential erosion did not occur. Instead, all three HPCs declined simultaneously as *fie* left active use – a pattern consistent with uniform community attrition rather than mechanism-specific erosion. The end state (prosodic shell without expressive force) is real, but the *route* to that end state was not the one the framework predicted.

The HPC-specific prediction is that the *order* of erosion tracks which mechanisms are disrupted: tight diagnostics (those maintained by fewer mechanisms) erode before loose ones (those maintained by multiple mechanisms). The first two cases (morphology gain, propositional content gain) are consistent with this: different perturbations produce different erosion orders, tracking which mechanism

[2] The analysis plan was committed to the project repository before data extraction (e18675b, 15 March 2026). The full pre-registration, data, and code are archived at https://github.com/BrettRey/interjections-hpc.

is disrupted. The third case (obsolescence) is not: the three properties declined together rather than sequentially, suggesting that community attrition affects all mechanisms simultaneously rather than picking them off in order. This is an informative failure – it identifies a boundary condition on the ordering prediction (it holds when a single mechanism is disrupted, not when the entire community of users contracts). The ordering prediction remains unavailable to prototype theory, which has no mechanisms, and to construction grammar, which has no perturbation probe.

To avoid immunizing the theory against disconfirmation, a valid perturbation test of the ordering prediction must meet three conditions: (i) the perturbation targets a single identified mechanism, not the entire community of users; (ii) the targeted mechanism is independently identifiable from the stabiliser inventory, not inferred from the erosion pattern itself; and (iii) the predicted ordering follows from the mechanism inventory before the data are examined, not fitted afterwards. The *fie* case violated condition (i): community attrition targeted all mechanisms simultaneously. The morphological gain cases (*wow* → *wowed*) satisfy all three conditions: gaining morphology disrupts prosodic-syntactic coupling (the confirmed stabiliser), predicting interjection$_{\text{syn}}$ erosion before interjection$_{\text{sem}}$. A diachronic test would require dated attestations for the verbal derivatives. Until such a test is conducted, the ordering prediction is an open commitment of the HPC account, not an established result – and if it fails under a valid single-mechanism perturbation, the case for HPC over SPC (Section 3) collapses to stable clustering without homeostatic maintenance.

4.4 Stability through turnover

The interjection cluster is stable even as its membership changes: individual members enter and leave, but the cluster persists.

The Corpus of Historical American English (COHA) illustrates this turnover. Among words tagged as interjections, *nay* ranks in the top 10 in the 1820s but falls to negligible frequency by the 2010s; *yeah*, entirely absent from the 1820s data, is among the five most frequent interjection-tagged forms by the 2010s.[3] The items change; the category doesn't.

Regional variation confirms the same point from a geographic angle. The GloWbE corpus (Davies & Fuchs, 2015, 1.9 billion words across 20 English-speaking countries) shows interjections distributed unevenly across varieties – *lah* almost exclusively in Singapore and Malaysia, *yaar* in India and Pakistan, *haba* in Nigeria – but the cluster profile (prosodic isolation, non-referentiality, emotive force, syntactic non-integration) is recognizable across all of them.

The cluster is also continuously replenished via onomatopoeia. Nonce interjections can be freely coined (Quirk et al., 1985, p. 74), and new coinages immediately adopt the cluster profile: they're prosodically isolated, non-referential, emotively laden, and syntactically non-integrated.

This generates a testable prediction: a newly conventionalized interjection should exhibit the full cluster profile without having traversed any of the recruitment pathways. Recent forms like *bruh*, *yeet*, and *oof* bear this out – all are prosodically isolated, non-inflecting, non-referential, and syntactically non-integrated, despite entering the language through different routes (phonological reduction, coinage, onomatopoeia respectively). The cluster persists even as its members turn over.

Dingemanse (2020) characterizes such items as LIMINAL SIGNS – signs at the boundary between sound and speech, whose liminality is functional, not defective. Cultural evolutionary processes

[3] Based on part-of-speech–tagged frequencies in COHA (Davies, 2010). Frequencies are normalized per million words. The interjection tag captures a conventional inventory, not a theoretically motivated one; the point here is the turnover pattern, not the tagging decisions.

"sample the landscape of possible resources". HPC provides the framework for that landscape: a property space with convergent maintenance by the stabilisers above.

5 Projectibility: what the category lets you predict

Section 2 showed what each property, individually, lets you infer. Section 4 identified the mechanisms that couple those properties. This section asks a different question: what does the package license, and for whom? The answer is the paper's central claim: interjections are a *pragmatic kind* whose richest projectibility concerns stance expression, interactional distribution, and conversational management, not truth-conditional denotation. The subsections build the case: partial-observation inference shows the mechanism works (§5.1); field-relative projectibility shows it works differently for different disciplines (§5.2); and the pragmatic-kind diagnosis (§5.4) identifies what makes interjections distinctive in the landscape of linguistic categories. From here on, bare "interjection" refers to the intersection of the three HPCs unless a subscript specifies otherwise.

5.1 Partial-observation inference

A circular account of projectibility says: if X is an interjection, it has interjection properties. That's trivially true of any category. HPC projectibility is different. It says: observing some cluster properties in a novel form licenses inference to other properties you haven't yet observed.

Consider *bruh*, a recently conventionalized form. A speaker encountering it for the first time observes that it's prosodically isolated and carries stance meaning. From those two observations alone, the three interjection HPCs license a battery of further predictions:

(4)
a. Interjection$_{\text{syn}}$: it won't inflect (**bruhs*, **bruhed*), will resist syntactic integration (**I think that bruh, we're late*), and won't take a determiner or function as an argument (**the bruh*, **Bruh surprised me*).
b. Interjection$_{\text{int}}$: it will be available for turn-initial or standalone use.
c. Interjection$_{\text{sem}}$: its expressive meaning will be use-conditional and speaker-oriented.
d. Diachronic: its etymological source is a content word (*brother*) that has undergone phonological reduction and semantic bleaching.

These predictions vary in strength, and the asymmetry matters:
- Near-entailments (a): the interjection$_{\text{syn}}$ predictions follow from the observed properties themselves (non-inflection, non-referentiality), not from HPC. Any framework recognizing these properties generates the same predictions. These are real but not HPC-specific.
- Cross-domain inferences (b, c): the interjection$_{\text{int}}$ prediction (interactional distribution from phonological form) and the interjection$_{\text{sem}}$ prediction (use-conditional meaning from stance expression) involve genuine leaps across linguistic levels. These are HPC-specific: they rely on the cluster's internal coupling, not on any single property.
- Expert-only predictions (d): the diachronic prediction (etymological trajectory from synchronic properties) is available only to linguists who recognize the category, not to naive speakers. This is projectibility for the discipline, not for the language user. That asymmetry is legitimate under Boyd (1999, p. 150)'s field-relativity, but it should be visible, not hidden.

All four succeed because the cluster's internal connections are maintained by identifiable processes: the non-referentiality ratchet (Section 4.2) connects loss of referential content to loss of argument structure to syntactic independence; the prosodic-syntactic coupling connects syntactic independence to prosodic isolation. The inferences are grounded in mechanisms, not in definitions.

5.2 Field-relative projectibility

Different fields track categories with roughly the same extension but different mechanisms and different inferential payoffs. Following Boyd (1999, p. 150) on the discipline-relativity of natural kinds, the three interjection HPCs introduced in Section 3.4 pay off here.

Interjection$_{syn}$ projects syntactic behaviour. If a novel form is categorized as interjection$_{syn}$, a syntactician can predict: supplement function, resistance to *that*-complementation, resistance to coordination with non-interjections, no argument structure (or at most a bare NP complement, as with *damn*). These are distributional predictions grounded in the prosodic-syntactic coupling mechanism from Section 4.2.

Interjection$_{sem}$ projects meaning type. If a form is categorized as interjection$_{sem}$ – overlapping with what Potts (2007) calls EXPRESSIVES – a semanticist can predict: use-conditional rather than truth-conditional contribution, speaker-oriented attitude (perspective dependence), nondisplaceability (the attitude is predicated of the utterance situation), and descriptive ineffability (Potts, 2007, p. 166). Potts defines expressives as a semantic type in a multi-dimensional logic; I treat these four properties as a cluster sustained by the bleaching mechanism from Section 4.1, not as a stipulative definition. The HPC reading is supported by graded membership: *damn* in *the damn car* has perspective dependence and use-conditionality but is arguably displaceable in some reported-speech contexts – it has some but not all of the cluster properties. This is the HPC signature, not a problem for the analysis. The extension differs from interjection$_{syn}$: attributive *damn* is interjection$_{sem}$ but not interjection$_{syn}$ – Potts (2007) calls it an expressive attributive adjective. Standalone *damn!* is both. The same lexeme belongs to different categories depending on construction, which is exactly the pattern the subscript architecture is designed to capture. Independent support comes from Gutzmann (2019), who argues that expressivity is a syntactic feature with agreement and movement reflexes – not merely a pragmatic overlay. His case studies of expressive adjectives, intensifiers, and vocatives show that the properties tracked by interjection$_{sem}$ have grammatical structure, reinforcing the claim that interjection$_{sem}$ is a genuine HPC, not a pragmatic residue.

Interjection$_{int}$ projects interactional distribution. If a form is categorized as interjection$_{int}$ – overlapping with Stivers (2019)'s response tokens and Dingemanse (2020)'s liminal signs – a conversation analyst can predict: turn-initial or standalone positioning, own intonation contour, stance display rather than propositional content, and availability for backchanneling. These are interactional predictions grounded in conversational routines and possibly in social indexing. The extension differs from interjection$_{syn}$: *mm-hm* is interjection$_{int}$ (a backchannel) but its status as interjection$_{syn}$ is debated (O'Connell & Kowal, 2005).

These are not three descriptions of the same category. They are three HPCs with overlapping but non-identical extensions, maintained by different mechanisms, projectible for different purposes. The overlap is explained by the fact that some mechanisms (prosodic isolation, non-referentiality) contribute to multiple HPCs simultaneously. The non-overlap is explained by the fact that other mechanisms are field-specific: structural analogy (the extension of existing syntactic patterns to new members by formal similarity) maintains interjection$_{syn}$ but not interjection$_{int}$; conversational routines maintain interjection$_{int}$ but not interjection$_{syn}$. A syntactician and a conversation analyst can carve the same words differently without either being wrong.

The architecture generates a cross-HPC prediction: because some mechanisms contribute to all three HPCs simultaneously, membership in one should predict membership in the others. A novel form categorized as $\text{interjection}_{\text{syn}}$ (prosodically isolated, non-inflecting, syntactically non-integrated) should also tend to be $\text{interjection}_{\text{sem}}$ (use-conditional, speaker-oriented) and $\text{interjection}_{\text{int}}$ (turn-initial, available for backchanneling). This is the default, and it holds for most interjections (*wow*, *ouch*, *hey*). The interesting cases are where the prediction fails: attributive *damn* is $\text{interjection}_{\text{sem}}$ but not $\text{interjection}_{\text{syn}}$; *mm-hm* is $\text{interjection}_{\text{int}}$ but debatably $\text{interjection}_{\text{syn}}$. Each failure identifies a point where the mechanisms decouple – and the pattern of decoupling is itself predicted by which mechanisms are field-specific.

The decomposition also isn't just notational. It generates predictions that the individual frameworks can't produce in isolation. The three constructions of *damn* illustrate this. For standalone *damn!*, all three HPCs agree: the form is prosodically isolated, syntactically non-integrated, use-conditional, and available as a stance marker. For *damn these mosquitoes!*, the picture is mixed: $\text{interjection}_{\text{syn}}$ holds (it heads an IntjP) but with residual verb-like complement-taking; $\text{interjection}_{\text{sem}}$ holds (the meaning is expressive); $\text{interjection}_{\text{int}}$ is weakened (it's not a backchannel or turn-management device). For attributive *the damn car*, they diverge further: $\text{interjection}_{\text{syn}}$ fails entirely (the form is syntactically integrated as an adjective), but $\text{interjection}_{\text{sem}}$ holds (use-conditional, perspective- dependent). Three constructions, three different patterns of convergence and divergence. Neither *CGEL*'s syntax nor Potts's semantics alone predicts this gradient. The subscript architecture does, because it tracks which mechanisms are active in each construction.

5.3 Category vs. function

A question from the *CGEL* tradition sharpens the projectibility point: what does the category label "$\text{interjection}_{\text{syn}}$" add beyond the function label "supplement"? Supplements of any category – NPs, AdjPs, relative clauses – share prosodic isolation and syntactic non-integration. But $\text{interjection}_{\text{syn}}$ adds a package of predictions that no other category-in-supplement-function supports:

- Non-referentiality. Supplements can be referential (*a university professor, Dr. Brown, ...*); interjections characteristically can't.
- Emotive/interpersonal force. Not all supplements express stance; interjections characteristically do.
- Free coinage. No other category in supplement function can be freely coined via onomatopoeia.
- Non-inflection. NP supplements inflect freely; interjections don't.
- Diachronic trajectory. Knowing something is a supplement makes no prediction about its etymology; knowing it's $\text{interjection}_{\text{syn}}$ does (bleaching paths, phonological reduction, constrained trajectories) for the linguist tracking the category diachronically.

The function label tells you about syntactic integration. The category label tells you about inference – it licenses a package of predictions that the function label alone can't provide. This distinction matters beyond the *CGEL* tradition: any framework that distinguishes what kind of thing an expression is from what job it's doing in a particular clause will face the same question, and the HPC answer is the same. The category earns its keep by licensing inferences that the function doesn't.

5.4 Interjections as a pragmatic kind

The subscript architecture reveals something about HPC theory itself. Not all kinds project in the same domain. Nouns project morphosyntactically: they take determiners, inflect for number, fill argument slots. Categorizing something as a noun tells you almost nothing about its pragmatics – nouns aren't a pragmatic category. For interjections, the situation is reversed. The three HPCs overlap, and the richest projectibility belongs to interjection$_{\text{int}}$ – the interactional cluster. Dingemanse (2024) reaches the same conclusion from interactional data: the most frequent and functionally central interjections are not emotive exclamations but interactional tools – continuers, repair initiators, change-of-state tokens – whose primary projectibility is about sequential position and conversational management. The forms at the intersection of all three HPCs are what I call a PRAGMATIC KIND: their strongest inferences are about how they will be used, not what they will denote.

Why pragmatic? Because mechanisms generate predictions within the domains they operate in. The prosodic-syntactic coupling from Section 4.2 predicts prosodic and syntactic behaviour – not morphological paradigms or truth-conditional content. Social indexing predicts persistence in identity-marking communities – not inflectional regularity. The mechanisms that maintain the three interjection HPCs are prosodic, interactional, and social, so the predictions they underwrite are in those domains. This isn't a stipulation; it's a consequence of which mechanisms happen to converge. Noun projectibility is morphosyntactic because the maintaining mechanisms (agreement, structural analogy, discourse frequency) operate in morphosyntactic domains.

This connects to the content/procedural distinction. Interjections are procedural, not conceptual (Traugott & Dasher, 2002): their projectibility is about use, not denotation. The interjection$_{\text{sem}}$ (expressive) cluster is the semantic reflex: use-conditional meaning is the semantic signature of procedural content. The parallel with other domain-specific kinds confirms the pattern. ANIMACY projects grammatically (argument realization, reference tracking); COLOUR projects ecologically (edibility, toxicity) but not grammatically – no language has colour-based verb agreement (cf. Seifart, 2010). In each case, the domain of projectibility follows from the maintaining mechanisms.

A functionalist might object: if the richest projectibility is pragmatic, why keep interjection$_{\text{syn}}$ in the grammar at all? The answer is that interjection$_{\text{syn}}$, thin as it is, is a genuine HPC with its own stabilisers and its own perturbation probe (Section 4.3). Neutralize prosodic isolation and the form is reclassified. The syntactic category does real distributional work – it predicts supplement function, embedding resistance, and coordination restrictions – even if those predictions are less rich than the pragmatic ones. A thin syntactic HPC is still a syntactic HPC. Boyd's framework doesn't require all kinds to be equally thick; it requires that they support reliable inference, and interjection$_{\text{syn}}$ does.

6 Discussion

6.1 Empirical predictions: HPC vs. prototype

Section 5 asked what categorizing a novel form as an interjection lets you predict about that form. This section shifts from instance-level projectibility to category-level predictions: what does HPC predict about the interjection intersection as a whole that prototype theory doesn't?

Prototype theory accommodates the interjection data – graded membership, fuzzy boundaries, better-and-worse exemplars – and has been applied to lexical categories since Rosch (1975). HPC doesn't replace prototype theory's descriptive achievements. It adds something prototype theory can't provide: mechanism-derived predictions with named defeat conditions. Three are testable for interjections.

Asymmetric fraying. If items drift away from the interjection cluster, tight diagnostics (non-inflection, complement resistance) should erode before loose ones (prosodic isolation). *Defeat:* diagnostics erode in no principled order. The evidence already in hand is consistent with the prediction. *Damn* retains complement-taking ability (*damn these mosquitoes!*) but has lost subject and tense – the tighter verbal diagnostics eroded first, while the looser one (complement frame) persisted. *Come on* and *look* have lost everything, including complements, consistent with a trajectory where tight diagnostics go first and loose ones follow. The perturbation cases from Section 4.3 point in the same direction: *wow* > *wowed* gains morphology and loses interjection$_{syn}$ before losing interjection$_{sem}$. Three cases are preliminary, not statistical, but the ordering is consistent and the prediction is falsifiable. Prototype theory predicts no particular order of erosion.

Conditioning recovers structure. Apparent gradience at the interjection boundary (fillers, routine formulae) should decompose when conditioned on register, dialect, or community. *Defeat:* conditioned and pooled models perform identically. A proof-of-concept test uses GloWbE frequency data (Davies & Fuchs, 2015): for the 10 most frequent interjection-tagged items plus regional and boundary items, frequency-per-million was extracted by country (20 countries) by raw string search (not POS tags, since the tagger fails on regional items) and the coefficient of variation (CV = SD/mean) was computed across countries.

The results show three tiers of geographic conditioning. Regional interjections (*lah*, *yaar*, *haba*) are near-categorical: *lah* occurs at 9.35 per million in Malaysia and 6.72 in Singapore but near zero elsewhere; *yaar* at 2.88 in Pakistan and 2.33 in India but near zero elsewhere; *haba* at 8.78 in Nigeria but near zero elsewhere. Core interjections show intermediate but structured variation: *ah* has the highest CV among core items (0.71), with Singapore (60.42) and Jamaica (53.16) at eight times the rate of Kenya (7.62). Even broadly distributed items like *yeah* (CV 0.55) show a 6:1 ratio between the United States (103.78) and Tanzania (16.85). Only *yes* and *no* approach uniformity (CVs of 0.21–0.22). Boundary items confirm the pattern: the fillers *um* (CV 0.61) and *uh* (CV 0.76) and the routine formula *bye* (CV 0.61) all show structured geographic variation.

Conditioning on country recovers structure at every level of the interjection landscape. The apparent gradience between core interjections, fillers, and boundary items is not noise; it is structured by variety. A Bayesian multilevel model (negative binomial with crossed random effects for item and country) confirms that adding country as a grouping factor improves prediction over an item-only baseline: the country-level random-effect standard deviation is 0.44 (95% CrI: 0.28–0.65), and the conditioned model outperforms the unconditional model by 18 ELPD units (SE 13) on exact leave-one-out cross-validation. The evidence is directionally clear but modest; the CV table above does the heavier lifting. Full model specification and results are in the supplementary materials.[4] Two methodological caveats: GloWbE's part-of-speech tagger, trained on inner-circle English, fails to tag regional interjections (*yaar* and *haba* receive no interjection tag at all), which means the tagged frequencies underestimate regional items; and the *ha* data is noisy because the tagger captures laughter representations (*HA! HA!*) alongside genuine interjection uses.

Hub-node asymmetry. One property should be more central to cluster stability than others. Candidates: prosodic isolation or non-referentiality. Removing the hub should destabilize the cluster more than removing a peripheral property. *Defeat:* all properties contribute equally. The perturbation probe in Section 4.3 provides preliminary evidence. Onomatopoeic interjections like *pop* can be syntactically integrated into clause structure. When prosodic isolation is neutralized – as when *pop* in *pop went the cork* functions as an adverb rather than a standalone interjection (Meinard, 2015,

[4] Available at https://github.com/BrettRey/interjections-hpc.

p. 152) – reclassification is immediate. Restoring partial referentiality, by contrast, allows the form to hover between noun and interjection uses (as with *God*). This suggests prosodic isolation may be the hub, but the evidence is indirect.

Each prediction has a named defeat condition. Prototype theory generates none of them – it describes the gradience but doesn't predict the pattern of erosion, the trajectory of entry, or the structure hidden in apparent variation.

A framework-level defeat condition: if a feature-based account (e.g., [+PERIPHERAL, -ARGUMENT-TAKING, +EXPRESSIVE]) generates all the same predictions without requiring mechanisms or cluster maintenance, then HPC is a relabelling, not an explanation. The HPC-specific payoffs are the ordering prediction under perturbation and the field-relative decomposition; if both fail under further testing, the case for interjections as an HPC collapses to a descriptive property list.

6.2 Interjections as a thin-but-genuine HPC

Where do interjections sit on the lexical-category landscape? Not in the wastebasket. ADVERB is widely regarded as a heterogeneous residual class – a label with no shared mechanism and no convergence. Knowing something is an adverb tells you it isn't a noun, a verb, or an adjective, and almost nothing else.

Interjections are different. They aren't a robust skeleton category like noun_{syn} or verb_{syn} (full mechanism convergence, deep entrenchment, early acquisition). But they are a genuine thin HPC, comparable to $\text{adjective}_{\text{syn}}$. The parallel is worth sketching. $\text{Adjective}_{\text{syn}}$, like $\text{interjection}_{\text{syn}}$, has fewer stabilisers than noun_{syn} or verb_{syn}, graded boundaries (with adverbs, participles, nouns), and cross-linguistic instability as an independent category – many languages lack a distinct adjective class. When English adjectives gain nominal morphology (e.g., nominalization: *the poor*, *the impossible*), they lose gradability and predicative use before losing attributive use – the same tight-before-loose erosion pattern that interjections show. Both categories resist classical definition while supporting reliable inference; both are maintained by a small set of mechanisms rather than by the dense convergence that sustains noun_{syn} and verb_{syn}. The category is maintained, not merely labelled. $\text{Interjection}_{\text{syn}}$ is thin on its own; the richness comes from the intersection of three overlapping HPCs ($\text{interjection}_{\text{syn}}$, $\text{interjection}_{\text{sem}}$, $\text{interjection}_{\text{int}}$), as Section 5 showed.

6.3 Boundary disputes as mechanism overlap

The graded boundaries that embarrassed classical category theory – interjections vs. nouns, verbs, adverbs, fillers, routine formulae – are predictions of HPC, not anomalies to be explained away. Each boundary reflects partial mechanism overlap.

Routine formulae (*bye*, *hello*, *sorry*) share prosodic isolation and pragmatic function with interjections but add addressee-directedness and social predictability (Ameka, 1992, pp. 109–110). They're maintained by additional mechanisms (social convention, politeness norms) that prototypical interjections don't require. In HPC terms, they share enough stabilisers to cluster nearby but not enough to sit at the centre.

Fillers (*um*, *uh*) share prosodic distinctiveness and syntactic non-integration but differ in emphasis, pause distribution, and citation-introduction ability (O'Connell & Kowal, 2005). They occupy a neighbouring region of the property space, maintained by partially overlapping mechanisms (processing-pressure management rather than emotive expression).

The Ameka-Wilkins debate is reframed: routine formulae are partial members of the interjection cluster, not a separate category (Wilkins, 1992, p. 142) and not fully inside it either (Ameka, 1992, p. 109). Both analysts were right about the data. The disagreement was about the framework, and HPC provides the framework that accommodates both observations.

7 Conclusion

English interjections are a thin-but-genuine HPC: a property cluster maintained by both diachronic recruitment pathways (grammaticalization, pragmaticalization, nonce coinage) and synchronic stabilisers (prosodic-syntactic coupling and possibly social indexing) and the non-referentiality ratchet as a boundary enforcer.

The category that Jespersen dismissed and Huddleston & Pullum set aside turns out to be a strong test case for HPC theory. Three diachronically distinct processes converge on the same synchronic cluster. The perturbation probe suggests that the stabilisers are genuine: removing one produces predictable, asymmetric category shifts.

What does categorizing something as an interjection let you predict? That is the question the HPC framework requires, and the answer is the paper's central claim: primarily pragmatic inferences – about prosodic behaviour, interactional distribution, and stance expression – not truth-conditional denotation. These inferences are grounded in identifiable mechanisms (prosodic-syntactic coupling, the non-referentiality ratchet) and reliable across novel instances. The field-relative decomposition (interjection$_{\text{syn}}$, interjection$_{\text{sem}}$, interjection$_{\text{int}}$) shows that the same lexemes participate in distinct HPCs for different fields. The traditional label names the intersection – a pragmatic kind, thin syntactically but rich interactionally. If that pragmatic projectibility proves illusory, or if a feature-based account generates the same predictions without mechanisms, the HPC analysis collapses.

For historical pragmatics specifically, the payoff is this: interjection formation involves at least three distinct diachronic processes (grammaticalization via semantic bleaching, documented in depth; pragmaticalization via syntactic deactivation and nonce coinage via onomatopoeia, both proposed with synchronic evidence), and HPC theory explains why they converge on a recognizable category despite their different trajectories. Gehweiler (2008)'s analysis of *gee!* is a single case study; the present paper extends it to a general claim about how the interjection intersection is recruited and maintained, while acknowledging that only the first pathway has the corpus documentation that historical pragmatics demands. The mechanisms are diachronic; the stability they produce is the kind of thing historical pragmatics is for.

Of the three predictions examined in Section 6, conditioning recovers structure is now supported by GloWbE data: geographic conditioning decomposes apparent gradience across the interjection landscape. Hub-node asymmetry remains untested: prosodic isolation should destabilize the cluster more than other properties when removed, and the perturbation evidence is suggestive but indirect. If the hub-node prediction fails, the HPC claim is weakened but not refuted. If it holds, interjections join the growing list of linguistic categories whose structure is better explained by mechanisms than by definitions.

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I compiled empirical material for the Wikipedia article "English interjections", of which I am the primary author. This paper reanalyses that material through an HPC lens.

Supplementary materials (data and analysis code for the GloWbE conditioning test) are available at https://github.com/BrettRey/interjections-hpc.

I drafted this paper with the assistance of Claude (Anthropic). I have reviewed and revised all content and take full responsibility for the final text.

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A Supplementary materials: GloWbE conditioning analysis

A.1 Data

Frequency counts for 18 items were extracted from the GloWbE corpus (Davies & Fuchs, 2015): 1.9 billion words across 20 countries. Items include the 10 most frequent interjection-tagged forms (*yes*, *no*, *oh*, *yeah*, *hi*, *hey*, *wow*, *hello*, *ah*, *ha*), three regional interjections (*lah*, *yaar*, *haba*), the semi-regional *eh*, the boundary items *cheers* and *bye*, and the fillers *um* and *uh*.

Figure 1 displays raw frequency per million words by country for each item.

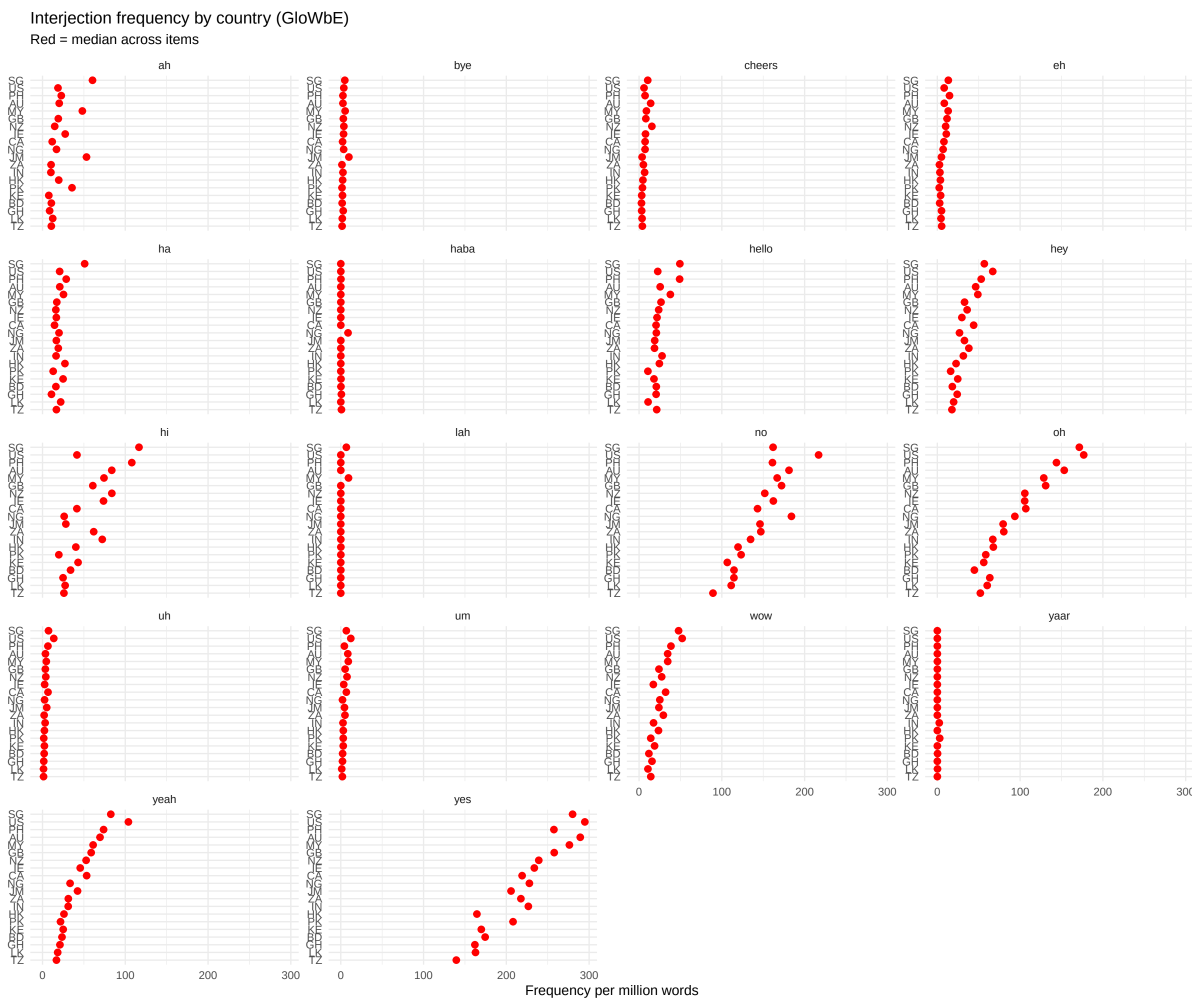

Figure 1: Interjection frequency per million words across 20 GloWbE countries. Red points indicate the median across items within each country. Regional items (*lah*, *yaar*, *haba*) cluster in one or two countries. Core items show structured cross-country variation. Only *yes* and *no* approach uniformity.

A.2 Model specification

Four Bayesian negative binomial models were fitted using `brms` 2.23 in R, with 4 chains of 2,000 iterations (1,000 warmup) each. The confirmatory models (m0–m2) use exact reloo for leave-one-out cross-validation; the exploratory model (m3) is assessed by PSIS diagnostics only.

- m0 (unconditional): `count ~ 1 + offset(log(words_m)) + (1 | item)`
- m1 (conditioned): `count ~ 1 + offset(log(words_m)) + (1 | item) + (1 | country)`
- m2 (item type): `count ~ 1 + item_type + offset(log(words_m)) + (1 | item) + (1 | country)`
- m3 (exploratory, country-by-type slopes): `count ~ 1 + item_type + offset(log(words_m)) + (1 | item) + (1 + item_type || country)`

Model m3 estimates country-level varying slopes by item type. Because several item-type cells are sparse (semi-regional: 1 item, boundary: 2, filler: 2, regional: 3), these slopes are thinly supported and m3 is treated as exploratory.

A.3 Results

Exact leave-one-out cross-validation (confirmatory models):

Model	ELPD diff	SE diff
m1 (conditioned)	0.0	0.0
m2 (item type)	−9.3	7.9
m0 (unconditional)	−17.9	13.3

The conditioned model (m1) is preferred. Adding item-type fixed effects (m2) does not improve on m1; the item random effect already captures between-item variation. The exploratory model (m3) is excluded from formal ranking because 9 observations retain Pareto $k > 0.7$ after moment matching, indicating unstable PSIS-LOO estimates.

Random effects from the conditioned model (m1):

- Country SD = 0.44 (95% CrI: 0.28–0.65)
- Item SD = 2.07 (95% CrI: 1.49–2.96)
- Ratio: 4.7:1 (item identity does more work, but country adds real structure)

The evidence is directionally consistent: conditioning on country improves prediction (ΔELPD = 17.9, SE 13.3), and the country random-effect SD is clearly non-zero. The SE is large relative to the point estimate, reflecting the small sample (18 items × 20 countries = 360 observations). The CV table in the main text provides the primary evidence for P5; the model confirms the pattern survives the crossed random-effects structure.

B Supplementary materials: obsolescence analysis (*fie* in COHA)

B.1 Data and coding

616 tokens of *fie* were extracted from COHA (1820–2019), excluding 141 OCR artefacts (where *fie* represents a misread of *he*, *the*, or *five*). Each token was coded for three binary properties: interjection$_{syn}$, interjection$_{sem}$, and interjection$_{int}$ (coding criteria in the pre-registration). The decade column was hidden during coding.

Intra-coder reliability (10% blind recode, $n = 62$):

Property	Agreement	Cohen's κ
interjection$_{syn}$	87.1%	0.68
interjection$_{sem}$	96.8%	0.90
interjection$_{int}$	96.8%	0.90

B.2 Raw proportions

Figure 2 shows the proportion of tokens retaining each property by decade. All three properties are near ceiling (85–100%) through the 1890s, decline sharply between 1900 and 1940, and fluctuate at low levels thereafter. The late rebounds (1980s, 2000s) reflect historical fiction and Shakespeare adaptations in which characters use *fie* as a live interjection.

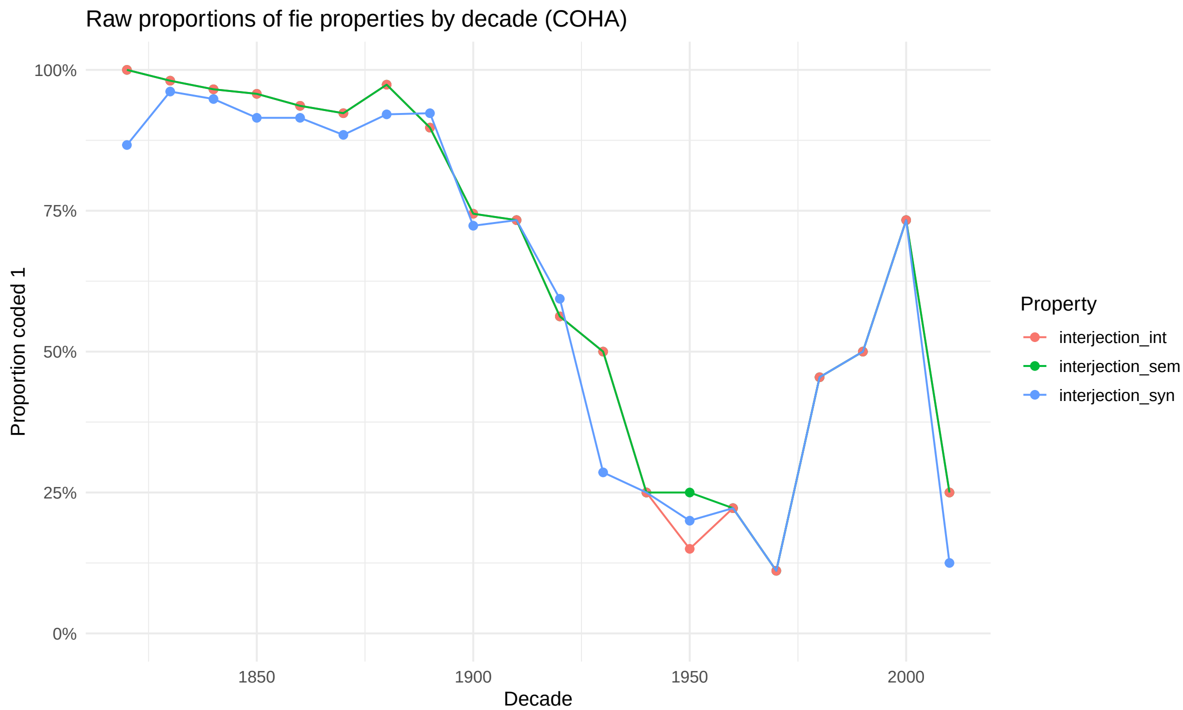

Figure 2: Proportion of *fie* tokens coded 1 for each property, by decade (COHA, 1820–2019). All three properties decline in parallel.

B.3 Confirmatory model

Bayesian logistic decline models (per the pre-registration) estimate the ED50 (τ: the year at which each property's probability crosses 0.5):

Property	τ (median)	95% CrI
interjection$_{syn}$	1947	1937–1958
interjection$_{int}$	1948	1939–1959
interjection$_{sem}$	1949	1941–1960

Pre-committed prediction: $\Pr(\tau_{\text{int}} < \tau_{\text{sem}} < \tau_{\text{syn}}) = 0.13$ (threshold for "inconsistent": < 0.25). The predicted sequential erosion was not observed. All three properties declined together. See Figure 3.

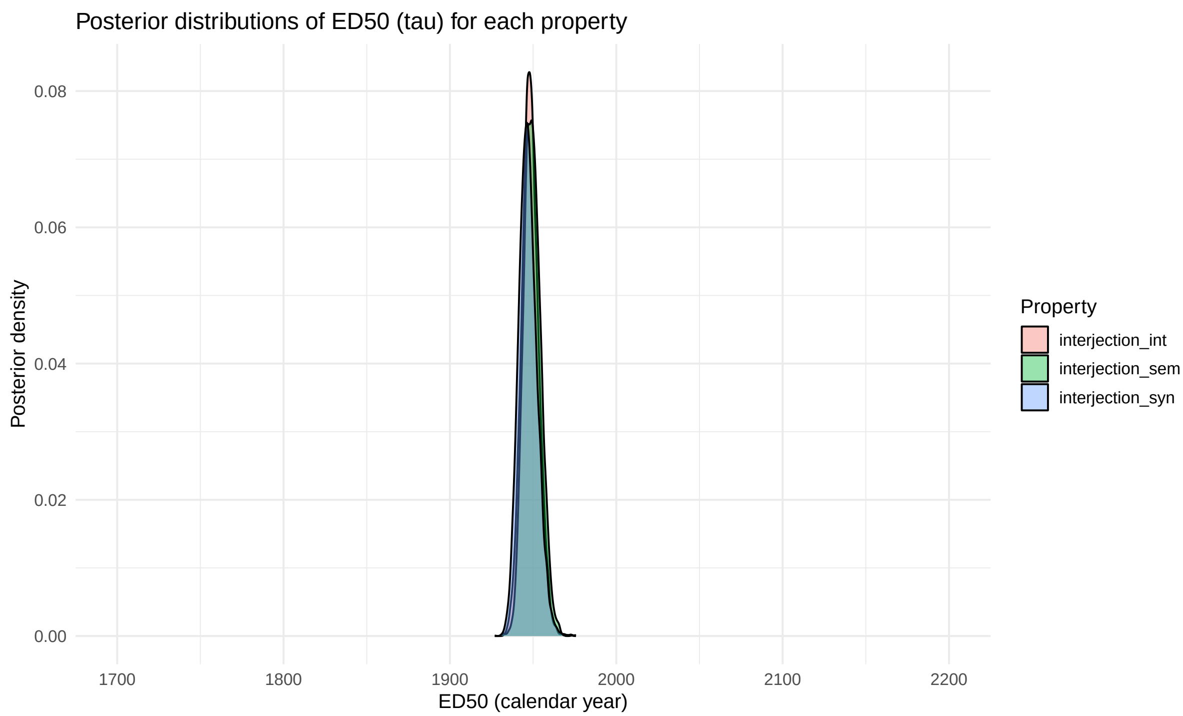

Figure 3: Posterior distributions of ED50 (τ) for each property. The three distributions overlap almost completely, indicating simultaneous decline.